Exp 10c: Design and Configuration of an Inter-network in the Lab Using Switch, Router and Ethernet Cables

AIM :

To design and configure an inter-network in the lab using switches, routers and Ethernet cables, and to show the IP address configuration of every network and device.

ALGORITHM :

1. Start the experiment.
2. Draw and label the design: two routers (Router0 and Router2), four switches and four LANs of five PCs each (two LANs on each router).
3. Connect the PCs to the switches and the switches to the routers using Ethernet (copper straight-through) cables, and connect the two routers to each other.
4. Configure the router interfaces with the gateway addresses of the LANs using the CLI and activate them with `no shutdown`.
5. Assign a static IP address, subnet mask and default gateway to every PC.
6. Verify the configuration of the routers (show ip interface brief) and of the PCs (ipconfig).
7. Test the connectivity by sending ICMP packets (ping) from the PCs to the router interface and to PCs of the other LAN of the same router, and observe the result in the PDU list.
8. Stop the experiment.

PROCEDURE :

The inter-network consists of four LANs (LAN 1 to LAN 4). A switch connects the PCs of a LAN and works inside that network, while a router connects different networks and forwards the packets between them. The router interface connected to a LAN is the default gateway of all the PCs of that LAN.

Step 1 – Cabling: Connect each PC to a FastEthernet port of its switch using a copper straight-through cable. Connect the switch of LAN 1 and the switch of LAN 2 to the two LAN interfaces of Router0, and the switch of LAN 3 and the switch of LAN 4 to the two LAN interfaces of Router2, using copper straight-through cables. Connect Router0 and Router2 to each other with a cable.

Step 2 – Configuring the routers (shown for Router0; Router2 is configured in the same way with 192.168.3.1 and 192.168.4.1):

```
Router>enable
Router#configure terminal
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
```

```
Router(config-if)#end
Router#show ip interface brief
```

Step 3 – Configuring the PCs: Select each PC → Desktop → IP Configuration → Static and enter the IP address, subnet mask and default gateway given in the table below. The switches work with their default settings and need no IP address.

IP address configuration:

LAN	Network	Switch	Router interface (gateway)	PC addresses (5 PCs)	Subnet mask
LAN 1	192.168.1.0	Switch3	Router0 Gig0/0 – 192.168.1.1	192.168.1.11 – 192.168.1.15	255.255.255.0
LAN 2	192.168.2.0	Switch4	Router0 Gig0/1 – 192.168.2.1	192.168.2.11 – 192.168.2.15	255.255.255.0
LAN 3	192.168.3.0	Switch1	Router2 Gig0/0 – 192.168.3.1	192.168.3.11 – 192.168.3.15	255.255.255.0
LAN 4	192.168.4.0	Switch5	Router2 Gig0/1 – 192.168.4.1	192.168.4.11 – 192.168.4.15	255.255.255.0

Step 4 – Testing: Open the Command Prompt of a PC and ping its gateway and a PC of the other LAN of the same router, or send a simple PDU between them. “Successful” replies in the PDU list show that the packets are delivered through the switches and routers.

OUTPUT :

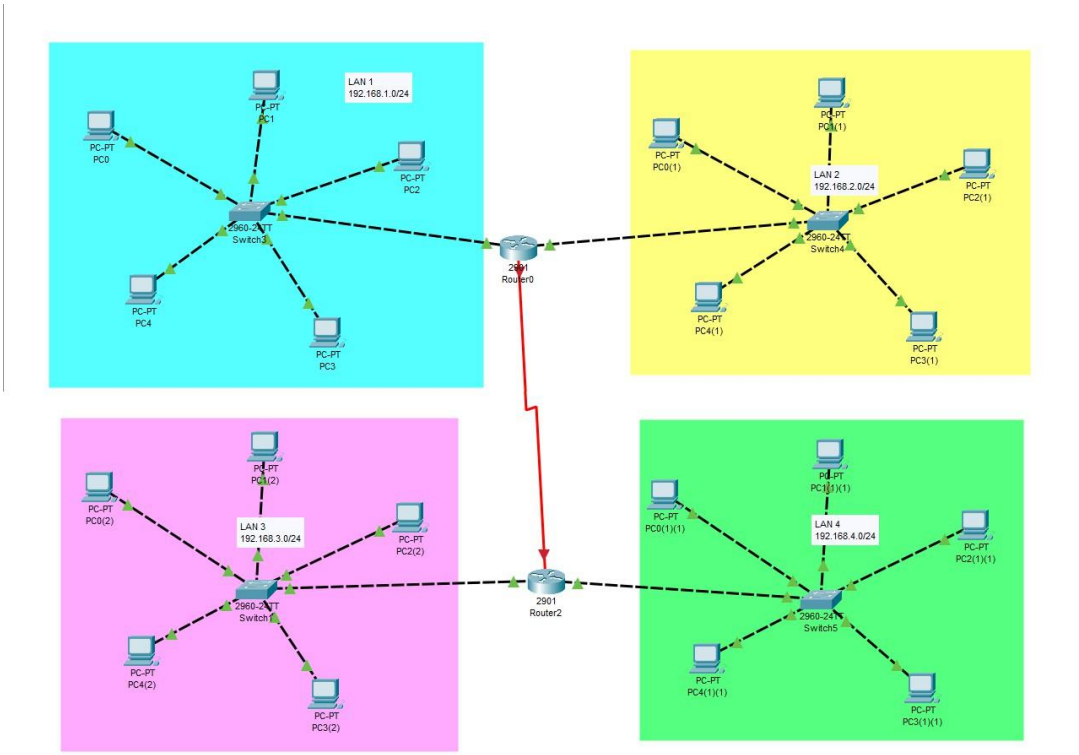


Fig 1. Designed inter-network – four LANs built with switches, connected through Router0 and Router2

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC4(2)	Router2	ICMP		0.000	N	12
	Successful	PC4(2)	PC2(1)(1)	ICMP		0.000	N	13
	Successful	PC0	Router0	ICMP		0.000	N	14

Fig 2. PDU list – ICMP packets transmitted successfully across the network

RESULT :

Thus, an inter-network was designed in the lab using switches, routers and Ethernet cables, the IP address configuration of the networks and devices was assigned and shown, and the packet transmission through the routers was successfully verified.